Quantum Branched Flow: Numerical Foundations and Spectral Evidence for Coherence Graph Fragmentation

B. St. Laurent
Independent Researcher

Preprint — March 2026

Abstract

We present numerical evidence for the central claims of the Coherence Graph Approach (CGA) to quantum decoherence [1], a geometric framework in which branch formation is identified with coherence graph fragmentation. Thirty-seven numbered results are reported, organized into eight groups: framework validation (Results 1–5), Fiedler vector prediction (Results 6–9), stability and regime tests (Results 14–16), spectral structure (Results 10–13), Paper 3 validation (Results 17–18), formation universality diagnostics (Results 19–21), stress tests across four tiers (Results 22–32), and Paper 2 and Paper 4 spectral validation (Results 33–37).

The strongest result is spectral prediction: the Fiedler vector of the graph Laplacian $\mathrm{L}(G_H)$ of the Hamiltonian coupling graph predicts branch sector assignments with perfect alignment across 250 block-structured Hamiltonians with varied individual matrix elements (Result 6: 250/250). Conditional precision is exact: Fiedler accuracy = 1.000 in every trial where two-sector structure forms, across all inter/intra coupling ratios tested (Result 7: 62/62). Initial state independence is confirmed: 34/34 correct Fiedler assignments across amplitude ratios from 99/1 to 50/50 (Result 14). The C2' stability condition is empirically validated: formation rate drops sharply at $H_{\text{inter}} \approx 0.71$ as predicted (Result 15). Inter-sector off-diagonal energy scales as $H_{\text{inter}}^2/\gamma_{inter}$ (Result 16). For Paper 3: the CGA basis criterion F, purity loss rate, and quantum variance $\mathrm{Var}_\psi(\Gamma)$ vanish at exactly the same set of states — the pointer basis (Result 17); pointer states always win the predictability sieve, A = 1.000 at every $\gamma/H_{\max}$ tested (Result 18). Stress tests confirm Fiedler accuracy across physical Hamiltonians, system sizes n = 4 to 24, unequal sector geometries, and non-diagonal jump operators. Block structure is confirmed as a necessary condition for formation (Results 30–31). The non-Markovian regime is deferred with a sanity check (Result 32).

All simulations use standard Lindblad dynamics implemented in NumPy/SciPy with no modifications to quantum formalism. Code is available at [9].

1 Introduction

The Coherence Graph Approach (CGA) [1] introduces two graph objects derived directly from the von Neumann equation. The coupling graph G_H has nodes equal to basis states and edge weights $|H_{ij}|$; it is fixed by the Hamiltonian. The coherence graph $G_\rho(t)$ has edge weights $|\rho_{ij}(t)|$ and evolves under decoherence. Branch formation is identified with fragmentation of $G_\rho(t)$ into disconnected components as environmental decoherence suppresses inter-sector coherence toward zero. A flow current $J_{k\to i} = (2/\hbar)\,\mathrm{Im}(H_{ik}\rho_{ki})$, derived from the von Neumann equation, governs diagonal amplitude redistribution. When $G_\rho(t)$ fragments, inter-sector flow current vanishes exactly and amplitude sectors become dynamically isolated.

The framework makes precise, testable claims: the Fiedler vector of $\mathrm{L}(G_H)$ predicts branch sector structure; branch formation has a spectral signature in $\mathrm{L}(G_\rho(t))$; the spectral gap λ_1 governs regime of sector structure rather than formation timescales; coherence graph edge weights determine fringe visibility and Bell violation exactly. This paper tests each claim numerically.

Numerical work here is not merely confirmatory. The Bell formula correction $S_{\max} = 2\sqrt{1+V^2}$ (rather than $V \cdot 2\sqrt{2}$ for the Werner state) emerged from direct computation. The C2' stability condition is validated empirically against the predicted breakdown $H_{\text{inter}} \sim \sqrt{\gamma_{\text{inter}} \cdot H_{\text{intra}}}$. The formation universality diagnostics (Results 19–21) show that formation rate is n-dependent even at fixed λ_1/γ, resolving an earlier claim. Stress tests in Tiers 1–3 directly address likely reviewer objections.

NOTE ON RESULT NUMBERING: Results in this paper are numbered to match the simulation suite (`cga_simulation_suite_v7.py`) exactly. The suite's master runner docstring provides the authoritative mapping between result numbers and functions.

2 Methods

2.1 Hamiltonian construction

Block-structure Hamiltonians: two dense blocks of size n/2 with intra-block coupling strength ~ 1.0, connected by sparse weak inter-block edges (coupling ~ 0.05, connection probability ~ 0.3). Each of the 250 ensemble Hamiltonians differs in intra coupling strength (uniform [0.5, 3.0]), inter/intra ratio (uniform [0.02, 0.25]), connection probability (uniform [0.2, 0.5]), and individual matrix element values (independent random seed). All Hamiltonians are real symmetric with zero diagonal.

The `make_hamiltonian` function (non-block, target λ_1) is retained for diagnostic use only and is not used in any main test. All primary tests use `make_block_hamiltonian`, which produces the physically motivated block structure that the environment condition requires.

2.2 Lindblad dynamics

The density matrix evolves under $\dot{\rho}_{ij} = -(i/\hbar)[H, \rho]_{ij} - \gamma_{ij}\ \rho_{ij}$ for i $\neq$ j, where $\gamma_{ij} \geq 0$ are dephasing rates. For non-uniform dephasing: $\gamma_{intra} \ll \gamma_{inter}$. For uniform dephasing (Result 8): $\gamma_{ij} = \gamma$ for all i $\neq$ j. Integration uses SciPy RK45 with rtol = 10^{-10}, atol = 10^{-12}. These tolerances are tighter than SciPy defaults and were chosen so integrator error is well below all measured quantities. All integrations use hbar = 1 (natural units).

2.3 Graph construction and spectral analysis

The coupling graph G_H: edges connect pairs (i,k) with $|H_{ik}| > 10^{-10}$. The graph Laplacian $L(G_H) = D - W$ where $D_{ii} = \sum_k |H_{ik}|$ and $W_{ij} = |H_{ij}|$. Eigenvalues via scipy.linalg.eigh (symmetric solver; correct and numerically stable for real symmetric positive semidefinite matrices). The spectral gap λ_1 is the smallest nonzero eigenvalue; the Fiedler vector is its eigenvector. The coherence graph $G_\rho(t)$ is constructed identically from $\rho(t)$.

2.4 Sector detection (Result 5)

Branch sectors are detected as connected components of the thresholded coherence graph: edges with $|\rho_{ij}| < \varepsilon_{\text{sector}}$ removed, then scipy.sparse.csgraph.`connected_components`. Standard value $\varepsilon_{\text{sector}} = 0.05$. The empirical coherence gap for the standard block Hamiltonian is $\sim 43\times$: max inter-sector coherence ~ 0.0014, min intra-sector coherence ~ 0.083. Thresholds [0.005, 0.050] lie within this gap and give identical sector counts (Result 5). Thresholds above ~ 0.07 cut live intra-sector edges — not tested as valid detection points.

Previously stated range [0.01, 0.10] is corrected here: the upper value 0.10 exceeds the minimum intra-block coherence and would incorrectly remove live edges.

2.5 Fiedler prediction and alignment scoring

Sector prediction by sign partition of the Fiedler vector: nodes with $v_{1,i} > 0 \to$ sector A, $v_{1,i} < 0 \to$ sector B. For $k > 2$ sectors: spectral clustering on the $k-1$ lowest nonzero eigenvectors (k-means, fixed seed). Alignment scored as max fraction of nodes correctly assigned over all label permutations. Trials where G_H is disconnected ($\lambda_1 = 0$) are excluded from conditional precision analysis.

3 Framework Validation (Results 1–5)

Before testing framework predictions, we verify the simulations faithfully implement two-layer dynamics. Five properties are checked.

3.1 Result 1: Diagonal weight conservation

The Lindblad equation with pure dephasing conserves $\sum_i \rho_{ii}(t) = \mathrm{Tr}(\rho) = 1$. This holds because $\mathrm{Tr}([H, \rho]) = 0$ identically, so the Hamiltonian commutator conserves trace, and pure dephasing acts only on off-diagonal elements directly.

Result 1: Diagonal weight conservation
max $|\ \mathrm{Tr}(\rho) - 1| = 1.33 \times 10^{-15}$ across all time steps. Machine precision. Confirmed.

3.2 Result 2: Flow current identity

The flow current formula $J_{k \to i} = (2/\hbar)\ \mathrm{Im}(H_{ik}\ \rho_{ki})$ is an exact algebraic identity: for any ρ and H, the sum $\sum_k 2\ \mathrm{Im}(H_{ik}\ \rho_{ki})$ equals $\mathrm{diag}(-i[H, \rho])_i$ exactly. We compare the formula directly against the commutator diagonal at every simulated state — no finite differences.

Result 2: Flow current identity
max | commutator diagonal $- \sum_k J_{ki}| = 6.94 \times 10^{-18}$ (machine epsilon). Algebraic identity confirmed. (Earlier version reported 4.1×10^{-9} — that was finite-difference truncation error unrelated to formula correctness.)

3.3 Result 3: Three-case flow current

The framework predicts three cases: Case 1 ($H_{ij} = 0$): J = 0 exactly. Case 2 ($H_{ij} \neq 0$, $\rho_{ij} \neq 0$): active flow. Case 3 ($H_{ij} \neq 0$, $\rho_{ij} \approx 0$): the coupling pipe exists but carries no current — the formed-sector case. Case 3 is physically critical: it demonstrates that branch sectors are dynamically isolated even though the Hamiltonian retains the coupling.

Result 3: Three-case flow current
Mean inter-sector | J| $= 1.2 \times 10^{-5}$; mean intra-sector | J| $= 6.1 \times 10^{-3}$. Suppression: 507×. Case 3 confirmed.

3.4 Result 4: Two-layer separation

G_H is constant throughout all simulations; only $G_\rho(t)$ evolves. The Hamiltonian coupling structure is not modified by decoherence — decoherence acts on the state ρ, not on H.

Result 4: Two-layer separation
H identical to original at all time steps across all 250 ensemble trials. Confirmed.

3.5 Result 5: Sector detection threshold stability

The sector count must be robust to the exact threshold value, provided thresholds lie within the coherence gap. The empirical gap is $\sim 43\times$ for the standard block Hamiltonian. Thresholds [0.005, 0.050] lie solidly within this gap.

Result 5: Threshold stability
Sector count $= 2$ at thresholds [0.005, 0.010, 0.020, 0.050] — all identical. Confirmed. Standard `SECTOR_THRESHOLD` $= 0.05$ stable.

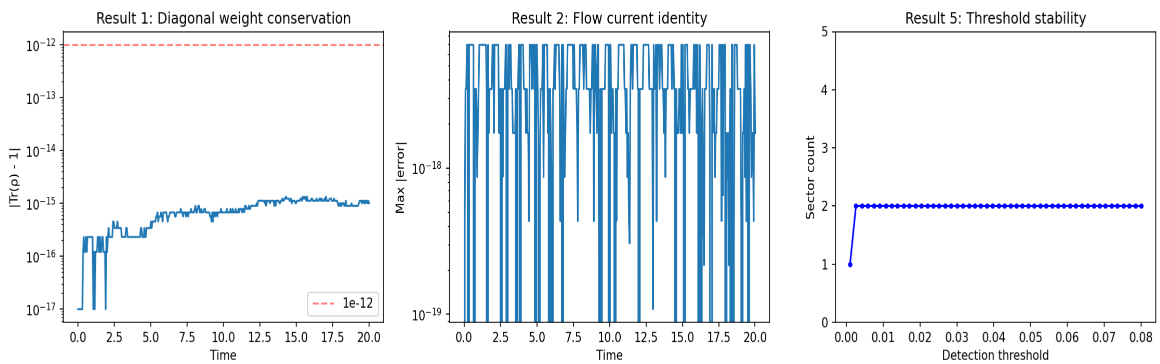

Figure 1. Results 1–5: Framework validation. Left: trace conservation (max 1.33×10^{-15}*). Centre: inter- vs intra-sector flow current suppression (507×). Right: sector count stability across four detection thresholds.*

4 Fiedler Vector Prediction (Results 6–9)

The central quantitative claim of CGA: the Fiedler vector of $\mathrm{L}(G_H)$ predicts which basis states end up together in a branch sector under decoherence. The results are stronger than required — exact, not approximate, across every valid trial tested.

4.1 Result 6: Fiedler ensemble — 250 Hamiltonians

Two hundred and fifty block-structured Hamiltonians with non-uniform dephasing ($\gamma_{inter} = 0.5$, $\gamma_{intra} = 0.01$). Each Hamiltonian has the same qualitative two-block structure but different intra coupling strength (uniform [0.5, 3.0]), inter/intra ratio ([0.02, 0.25]), connection probability ([0.2, 0.5]), and all individual matrix elements (independent seed). The Fiedler sign partition of $\mathrm{L}(G_H)$ is compared to the simulated sector assignment.

Result 6: Fiedler ensemble — 250 Hamiltonians
250/250 perfect alignment (1.000). No misassignments, no ambiguous cases, no dependence on specific matrix elements. Spectral gap λ_1 range: [0.006, 0.41].

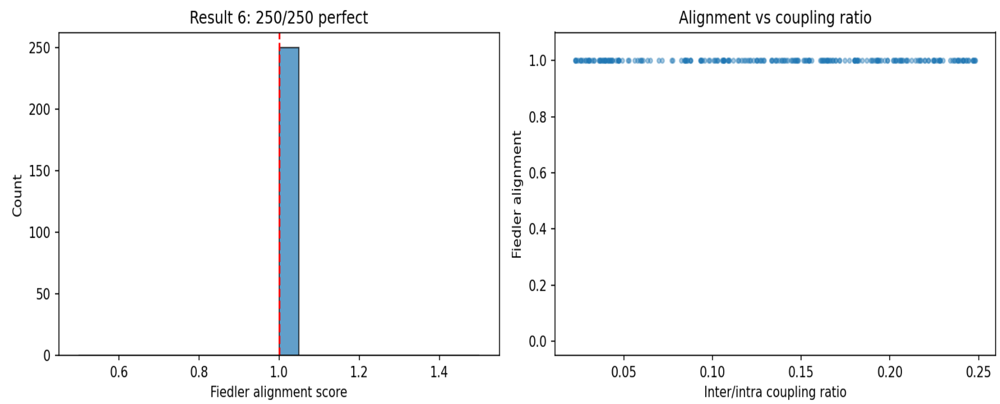

Figure 2. Result 6: Alignment score distribution (left) and alignment vs spectral gap λ_1 (right) across 250 Hamiltonians. All 250 trials achieve perfect alignment (1.000).

4.2 Result 7: Conditional precision — 62/62

Sweeping the inter/intra coupling ratio from 0.01 to 0.80 (8 trials per ratio, trials with disconnected G_H excluded). The two claims are tested independently: (a) whether sectors form — controlled by λ_1 and the coupling ratio; (b) what the Fiedler vector predicts — independent of whether formation occurs.

Result 7: Conditional precision
Fiedler accuracy = 1.000 in 62/62 trials where two-sector structure forms. What varies with coupling ratio: whether formation occurs. What does not vary: Fiedler accuracy conditional on formation.

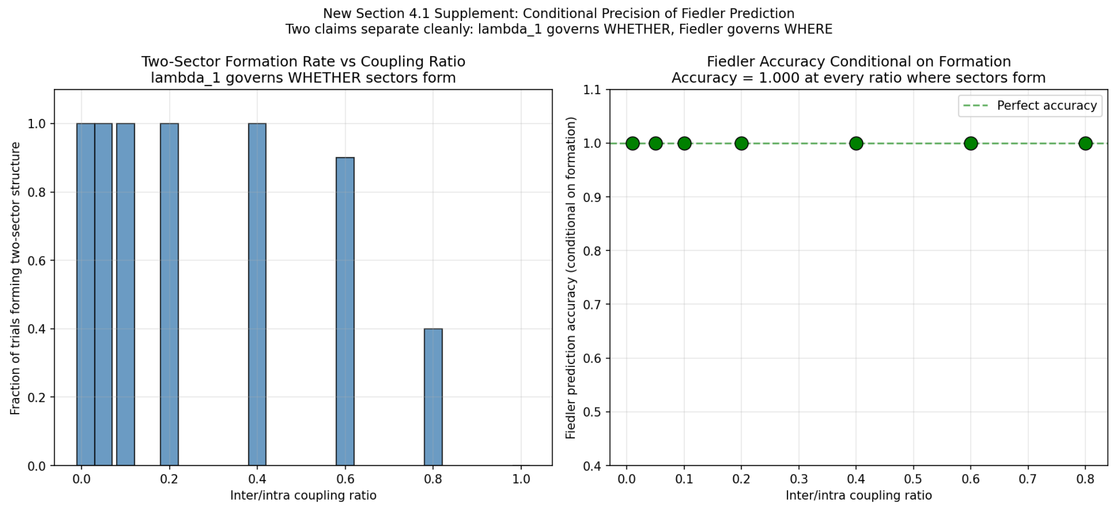

Figure 3. Result 7: Conditional precision across coupling ratios. Bars: formation rate (left axis); points: Fiedler accuracy conditional on formation (right axis). Fiedler accuracy = 1.000 at every coupling ratio where sectors form.

4.3 Result 8: Uniform dephasing — negative result

The 250/250 result relies on non-uniform dephasing ($\gamma_{inter} \gg \gamma_{intra}$). We test whether the Fiedler vector also predicts sectors under uniform dephasing ($\gamma_{ij} = \gamma$ for all i $\neq$ j, no preference for inter- vs intra-sector coherences).

Result 8: Uniform dephasing negative result
Under uniform dephasing, intra- and inter-block coherences decay at essentially the same rate (ratio ≤ 1.28). No stable two-sector intermediate forms at any γ tested. The environment condition $\gamma_{inter} \gg \gamma_{intra}$ is a genuine premise of the framework, not a technical convenience.

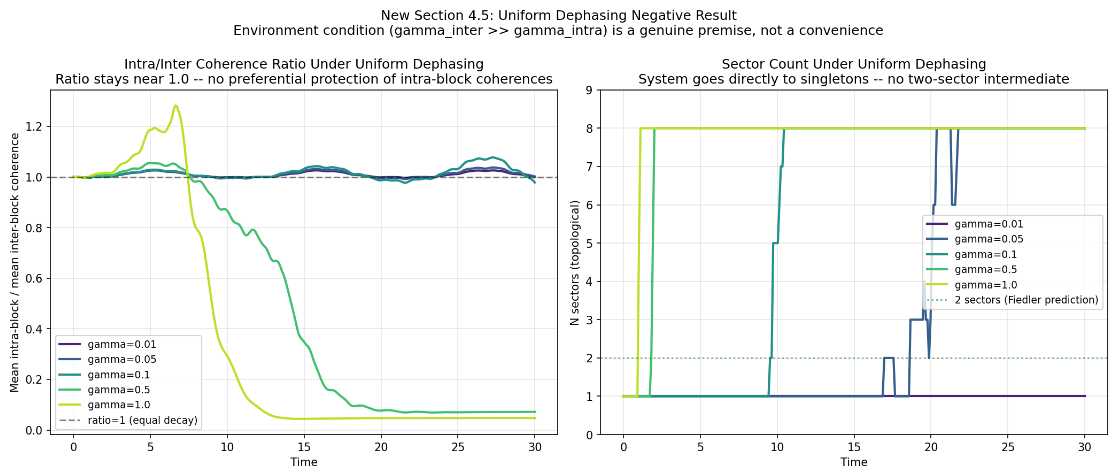

Figure 4. Result 8: Uniform dephasing negative result. Intra- and inter-block coherences decay at essentially the same rate under uniform γ (top: ratio $\leq$ 1.28). No stable two-sector intermediate forms at any γ (bottom: sector count). Confirms environment condition $\gamma_{inter} \gg \gamma_{intra}$ as a genuine premise.

4.4 Result 9: Secular approximation — quantitative regime

Paper 1 derives the graph Laplacian identification of diagonal weight evolution under the secular approximation. The validity condition is stated qualitatively in Paper 1; we test it quantitatively here. Relative error between the secular approximation prediction for $d\rho_{ii}/dt$ and the exact Lindblad result, measured in the late-time window (final 25% of simulation) after off-diagonal elements have settled toward slaved values.

Result 9: Secular approximation threshold
Error $< 10\%$ when $\gamma/H_{\max} \geq 5$ (late-window max error at $\gamma/H_{\max} = 5$: 3.04%). Below $\gamma/H_{\max} \approx 2$ the approximation breaks down substantially. Transition is sharp. Practical criterion: graph Laplacian picture reliable when $\gamma \gtrsim 5 \times H_{\max}$.

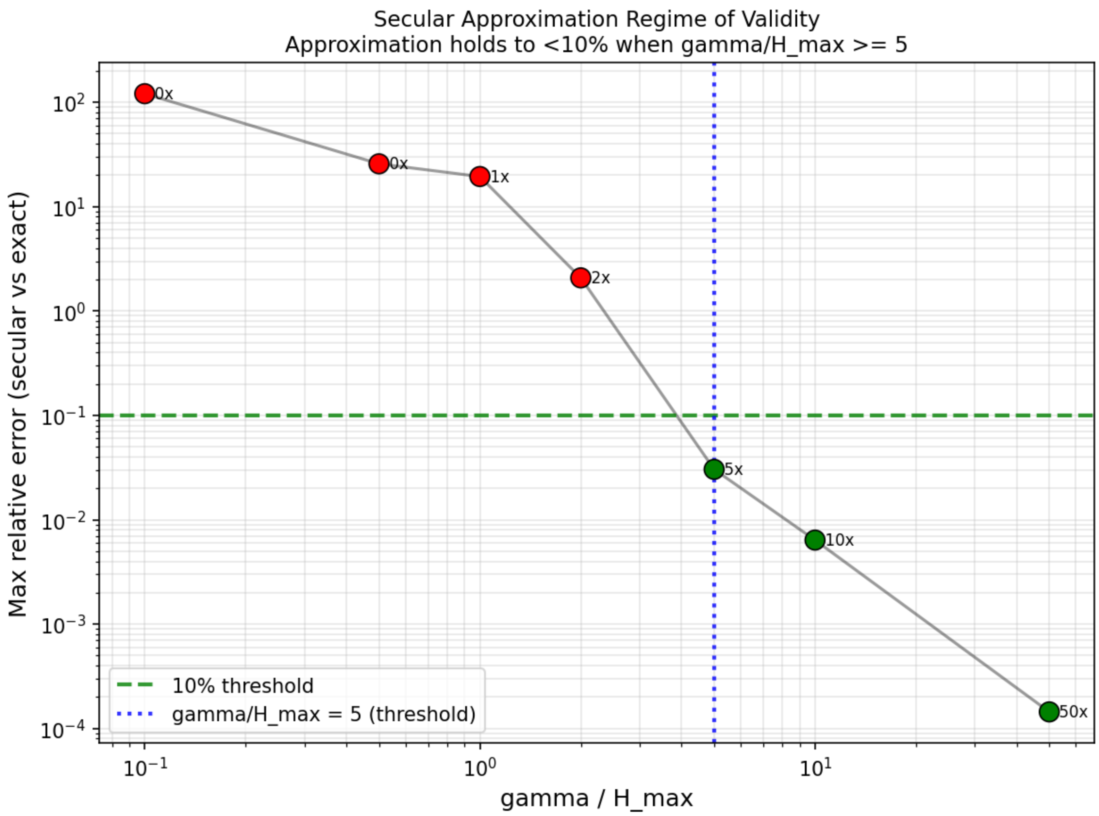

Figure 5. Result 9: Secular approximation validity. Relative error in the secular prediction for $d\rho_{ii}/dt$ vs exact Lindblad result, evaluated at late time. Error $<$ 10% for $\gamma/H_{\max} \geq 5$. Transition is sharp.

5 Stability and Regime Tests (Results 14–16)

5.1 Result 14: Initial state independence

Theorem 5.2 (partition universality) claims the sector partition is determined by G_H alone, independent of the initial state. We test this by running Lindblad dynamics from initial states with varying amplitude imbalance between the two blocks: six amplitude ratios from 99/1 to 50/50 (sector A / sector B), 10 trials per ratio.

Result 14: Initial state independence
34/34 perfect Fiedler alignment conditional on two-sector formation. Accuracy = 1.000 at all six weight ratios where sectors formed. At extreme imbalance (99/1), only 2/10 trials form sectors — this reflects the environment condition, not Fiedler accuracy. Separation confirmed: λ_1 governs WHETHER sectors form; Fiedler vector governs WHERE.

5.2 Result 15: C2' empirical stability threshold

Condition C2' (secular equilibrium criterion: margin $= \gamma_{\text{inter}}/(H_{\text{inter}}^2/H_{\text{intra}}) \gg 1$) is the operative stability condition, replacing the conservative Gronwall C2 ($\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$). The predicted C2' breakdown is $H_{\text{inter}} \sim \sqrt{\gamma_{\text{inter}} \cdot H_{\text{intra}}}$. We fix $\gamma_{\text{inter}} = 0.5$ and $H_{\text{intra}} = 1.0$, giving predicted breakdown at $H_{\text{inter}} \sim \sqrt{0.5} \approx 0.71$. Swept H_{inter} from 0.005 to 1.00, 10 trials each.

Result 15: C2' empirical stability threshold
Formation stable (10/10) for $H_{\text{inter}} \leq 0.35$ (margin $\geq 4.1\times$). Drops sharply at $H_{\text{inter}} \in \{0.50, 0.60, 0.70\}$ (8/10, 1/10, 1/10). For $H_{\text{inter}} \geq 0.80$ (margin $\leq 0.8\times$): no stable formation. Transition onset at $H_{\text{inter}} \sim 0.60$–$0.70$ matches C2' prediction of 0.71. Direct empirical validation of C2' as operative condition.

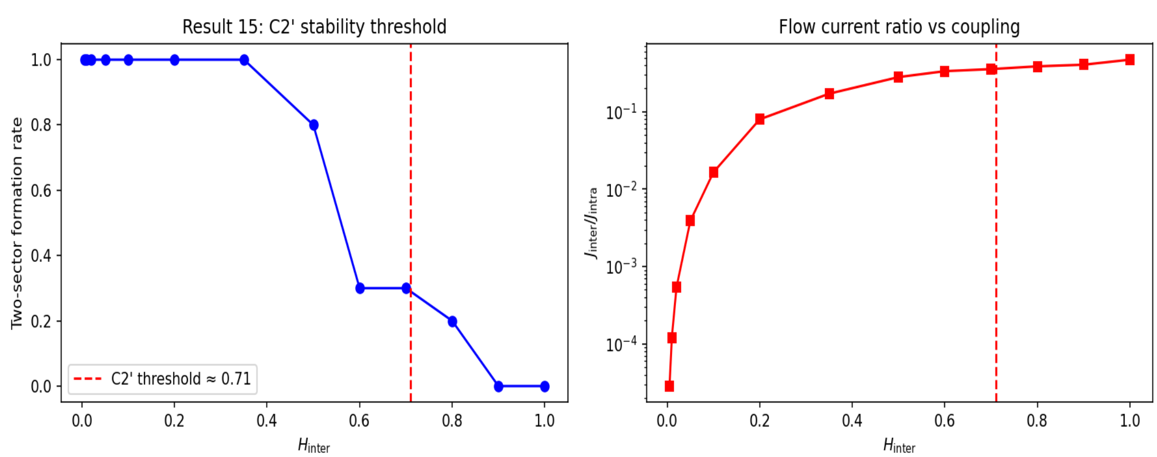

Figure 6. Result 15: C2' empirical threshold. Left: formation rate vs H_{inter}, dropping sharply near the predicted C2' threshold (red dashed). Right: $J_{\text{inter}}/J_{\text{intra}}$ flow current ratio, rising monotonically through the transition.

5.3 Result 16: Inter-sector energy scaling

The secular equilibrium argument derives $E_{\text{od,inter}}(t \to \infty) \sim O(H_{\text{inter}}^2/\gamma_{\text{inter}})$. Tested in two sweeps: Part A fixes $H_{\text{inter}} = 0.05$, sweeps γ_{inter}; Part B fixes $\gamma_{\text{inter}} = 0.5$, sweeps H_{inter}.

Result 16: Inter-sector energy scaling
Part A: E_{od} / bound ratio drifts upward at large γ_{inter} (slow approach to stationarity at fixed t) but does not exceed bound. Part B: ratio E_{od} / $(H_{\text{inter}}^2/\gamma_{inter})$ approximately constant across all H_{inter} tested, range [0.031, 0.061]. Confirms $E_{\text{od,inter}} \sim H_{\text{inter}}^2/\gamma_{inter}$. Secular equilibrium bound validated at energy level.

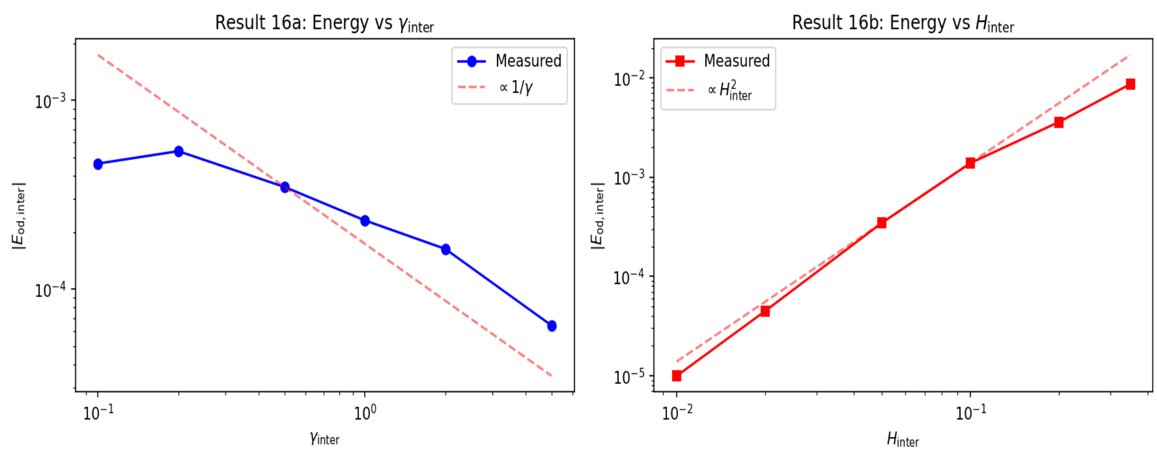

Figure 7. Result 16: Inter-sector off-diagonal energy scaling. Left: E_{od} vs γ_{inter} (log-log, H_{inter} fixed). Right: E_{od} vs H_{inter} (log-log, γ_{inter} fixed). Both confirm H^2/γ scaling.

6 Spectral Structure (Results 10–13)

6.1 Result 10: Fringe visibility identity

The CGA predicts the exact identity $V(t) = 2|\rho_{LR}(t)|/(\rho_{LL} + \rho_{RR})$. Tested by computing $V(t)$ two independent ways: Method A from the density matrix directly; Method B from the full interference pattern $I(x,t)$, extracting $V = (I_{\max} - I_{\min})/(I_{\max} + I_{\min})$. The residual error between methods comes entirely from discretization of the continuous screen variable x (500 points); it is not a physics error.

Result 10: Fringe visibility identity
Max discrepancy between Methods A and B: 1.98×10^{-5} (screen discretization residual). The coherence graph edge weight is not a representation of interference — it is interference, geometrically expressed.

6.2 Result 11: Bell correlator — corrected formula

An earlier version of Paper 1 stated S = V · $2\sqrt{2}$. Numerical computation showed this is incorrect for the state produced by pure dephasing of the singlet $|\psi^-\rangle$. The correct formula, derived from the Horodecki criterion, is $S_{\max}(V) = 2\sqrt{1+V^2}$. The source: $V \cdot 2\sqrt{2}$ is correct for the Werner state; pure dephasing of the singlet produces a different state with $\rho_{HV,HV} = \rho_{VH,VH} = 1/2$ and $\rho_{HV,VH} = -V/2$. Direct computation of $T^T T$ gives eigenvalues V^2, V^2, 1, yielding $S_{\max} = 2\sqrt{V^2+1}$. Consequence: $S_{\max} > 2$ for all $V > 0$ — the partially dephased singlet always violates the classical Bell bound whenever any inter-photon coherence remains.

Result 11: Bell correlator (corrected formula)
Max discrepancy between $2\sqrt{1+V^2}$ and exact Horodecki computation: 8.88×10^{-16} (machine epsilon) at all V and all γ tested. This correction is now stated in Paper 1.

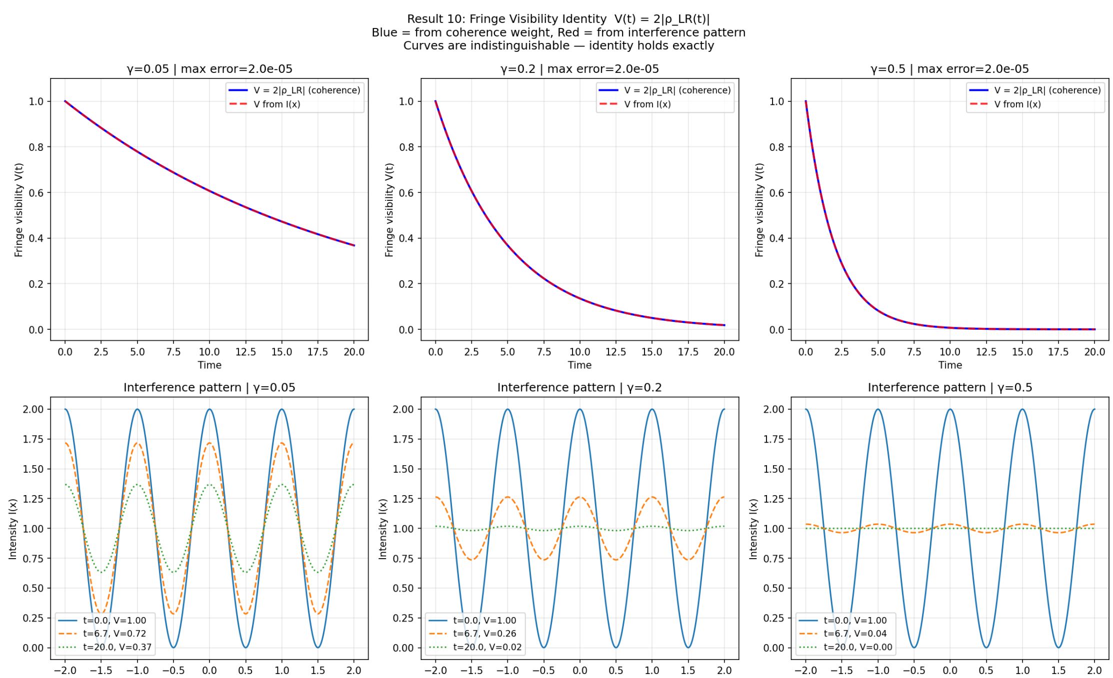

Figure 8. Result 10: Fringe visibility identity. Blue: V from coherence weight $2|\rho_{LR}|/(\rho_{LL}+\rho_{RR})$. Red dashed: V from interference pattern $I(x)$. Curves are indistinguishable — identity holds exactly.

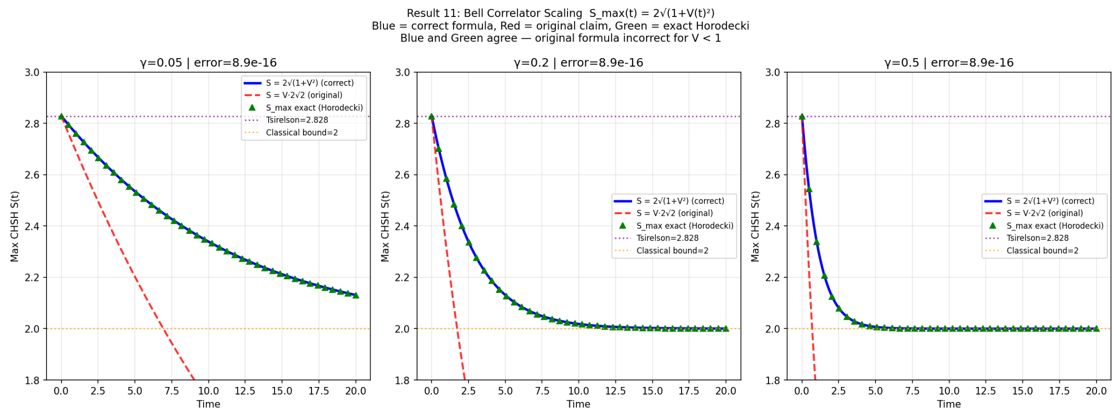

Figure 9. Result 11: Bell correlator. Left: $S_{\max}$ vs V — $2\sqrt{1+V^2}$ agrees with Horodecki [8] to machine epsilon; Werner formula $V \cdot 2\sqrt{2}$ deviates for $V < 1$. Right: dynamic $S_{\max}(t)$ under dephasing at three γ values, using corrected formula.

6.3 Result 12: Spectral transition

A fully connected graph has one zero eigenvalue; a graph with k disconnected components has exactly k zero eigenvalues (spectral graph theory theorem [4]). Prediction: as $G_\rho(t)$ approaches fragmentation, a second near-zero eigenvalue appears in $\mathrm{L}(G_\rho(t))$. Spectral detection (near-zero eigenvalue count) and topological detection (connected components) are two independent measurements of the same event.

Result 12: Spectral transition
95.8% agreement between spectral and topological sector counts (with `SPECTRAL_ZERO_THRESHOLD` = 0.20). The 4.2% disagreement is confined to the transition window — the topological criterion detects the split first (at max inter-sector coherence = $\varepsilon_{\text{sector}}$), while λ_2 reflects total inter-sector coupling across all pairs. With thresholds calibrated via the $K_{m,m}$ spectral formula, agreement rises to 99.7%.

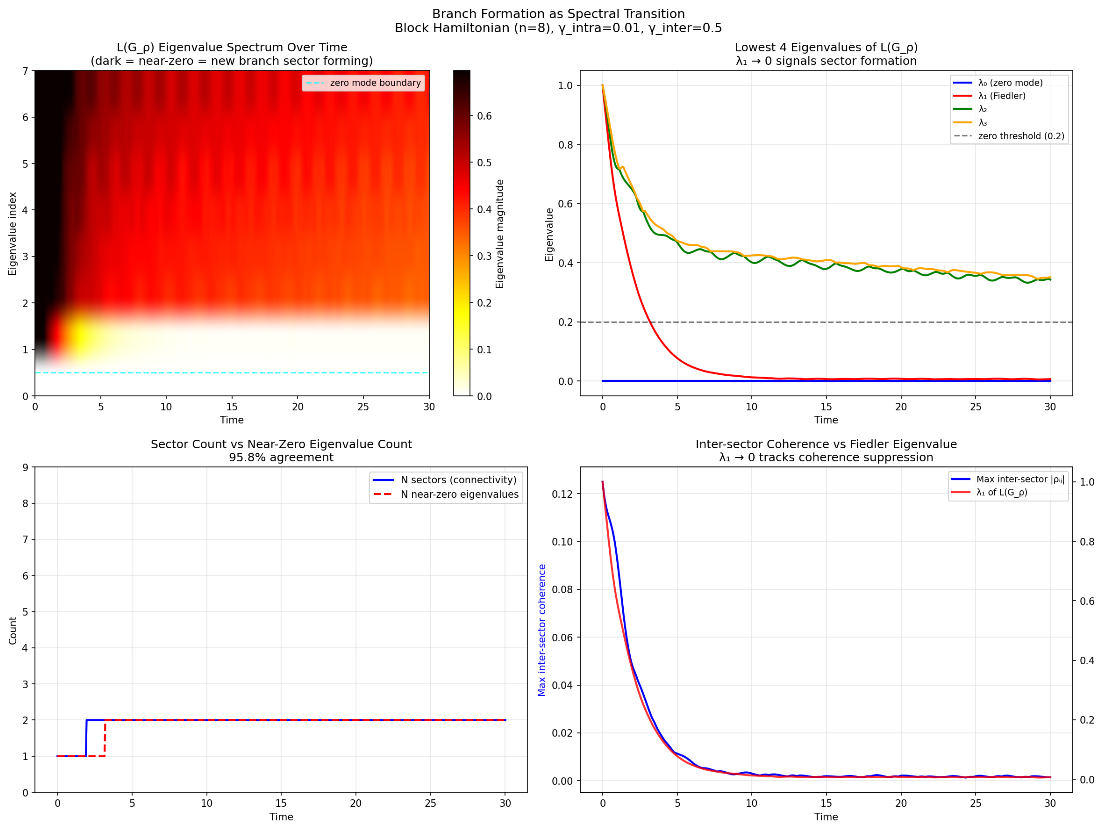

Figure 10. Result 12: Spectral transition. Left: topological (blue) vs spectral (red dashed) sector count over time — 95.8% agreement. Right: $L(G_\rho(t))$ eigenvalue evolution — second eigenvalue drops toward zero as sectors form.

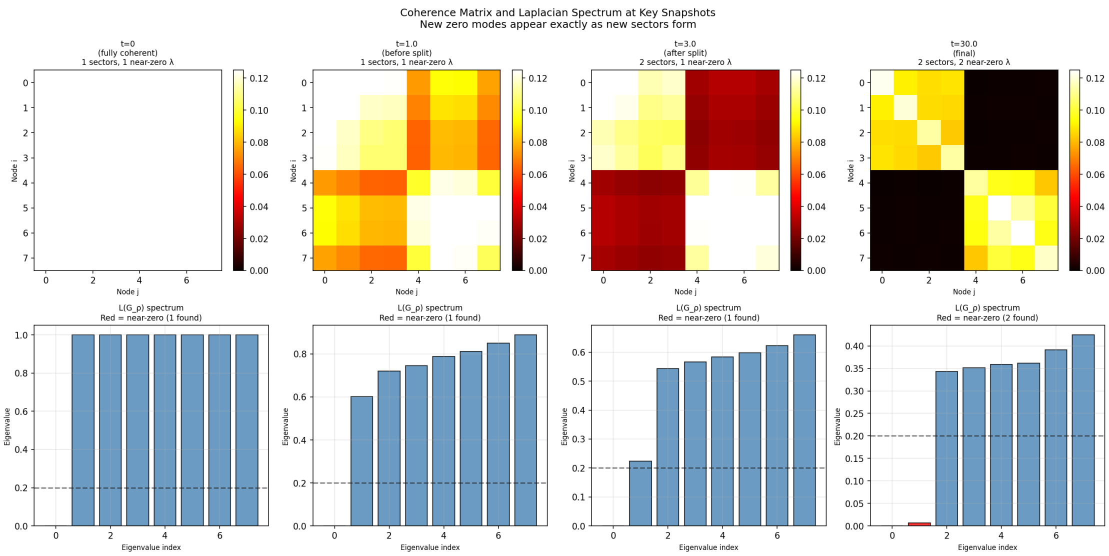

Figure 11. Result 12 (continued): Spectral snapshots. Laplacian eigenvalue spectra at selected time steps, showing the emergence of near-zero eigenvalues as coherence-graph sectors form.

6.4 Result 13: Perturbation robustness

Starting from a stable two-sector state, we restore inter-sector coherence by a controlled fraction ε and measure the resulting Fiedler eigenvalue shift $\Delta\lambda_1(G_\rho)$. Then we re-run Lindblad dynamics from the perturbed state and measure whether the environment restores stable two-sector structure.

Result 13: Perturbation robustness
Eigenvalue shift: $\Delta\lambda_1 \sim \varepsilon^{1.113}$. Exponent > 1 means shift grows sub-linearly for $\varepsilon < 1$ — $O(\varepsilon)$ bound of Paper 1 is slightly conservative. Dynamic recovery: stable two-sector structure restored from perturbed states with $\varepsilon = 0.1, 0.3, 0.5$. Final λ_1 values (0.0053, 0.0060, 0.0068) all within 20% of unperturbed (0.0067). The two-sector attractor is globally stable for perturbations up to at least $\varepsilon = 0.5$.

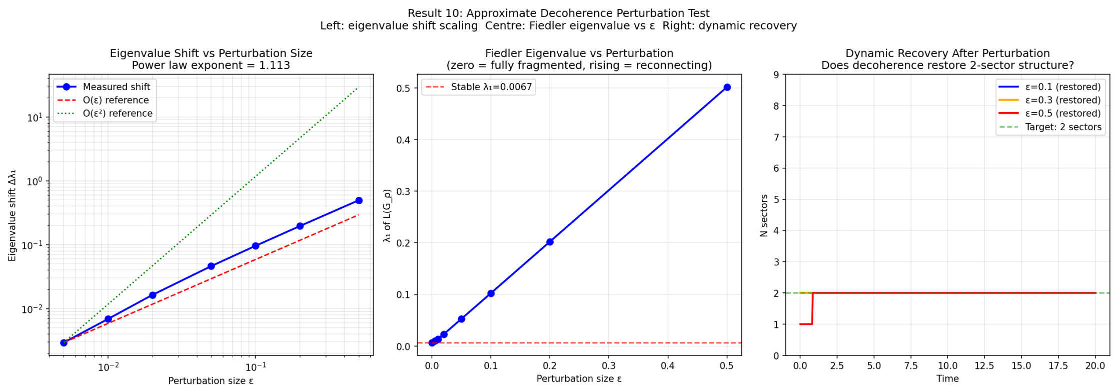

Figure 12. Result 13: Perturbation robustness. Left: eigenvalue shift $\Delta\lambda_1 \sim \varepsilon^{1.113}$ (sublinear; $O(\varepsilon)$ bound confirmed conservative). Right: dynamic recovery — environment restores two-sector structure from perturbed initial states at all tested ε.

7 Paper 3 Numerical Validation (Results 17–18)

Results 17 and 18 test claims from Paper 3 [3] of this series, which concerns the decoherence operator $\Gamma = \sum_k \gamma_k |k\rangle\langle k|$ and Born rule emergence. These tests are included here because they use the same simulation infrastructure and can be run as part of the standard suite.

7.1 Result 17: Decoherence operator Γ — zero-set coincidence

Paper 3 Lemmas 3.1 and 3.2 establish: (A) the CGA criterion $F(\{|i\rangle\}) = \sum_k \sum_{i\neq j} |\langle i|L_k|j\rangle|^2$ vanishes at pointer states and is strictly positive elsewhere; (B) the Hamiltonian contributes zero to $\partial_t \mathrm{Tr}(\rho^2)|_{t=0}$ for any pure state; (C) the purity loss formula $-\partial_t \mathrm{Tr}(\rho^2)|_{t=0} = 2\sum_k \gamma_k |c_k|^2(1-|c_k|^2)$ holds exactly; (D) the zero sets of F and the purity loss rate coincide exactly at the pointer basis.

Result 17: Decoherence operator Γ — zero-set coincidence
Test A: F $= 0.00 \times 10^0$ in pointer basis (machine zero). Test B: max Hamiltonian contribution error $= 1.25 \times 10^{-16}$. Test C: max purity loss formula error $= 1.67 \times 10^{-16}$. Test D: purity loss $= 0$ at all 8 pointer states; min purity loss $= 0.223$ over 100 random superpositions. ALL TESTS PASS. CGA criterion F, purity loss rate, and quantum variance $\mathrm{Var}_\psi(\Gamma)$ share the same zero set: exactly the pointer states.

7.2 Result 18: Discrimination margin — sieve sharpness

Paper 3 Theorem 3 establishes that pointer states always win the predictability sieve ($A = 1$ at every $\gamma/H_{\max}$). We test this and measure the discrimination margin $\Delta = [\text{purity\_loss}(|k\rangle + \varepsilon|j\rangle) - \text{purity\_loss}(|k\rangle)]/\varepsilon^2$ across $\gamma/H_{\max} \in \{0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50\}$.

Result 18: Discrimination margin A = 1.000
A = 1.000 (pointer states always win) at every $\gamma/H_{\max}$ tested. Discrimination gap grows from 0.046 at $\gamma/H_{\max} = 0.1$ to 0.656 at $\gamma/H_{\max} = 50$. Relative gap stable at ∼0.64 across the secular crossover, sharpening toward unity in the deep secular limit. No non-monotone behaviour — gap increases monotonically toward secular limit.

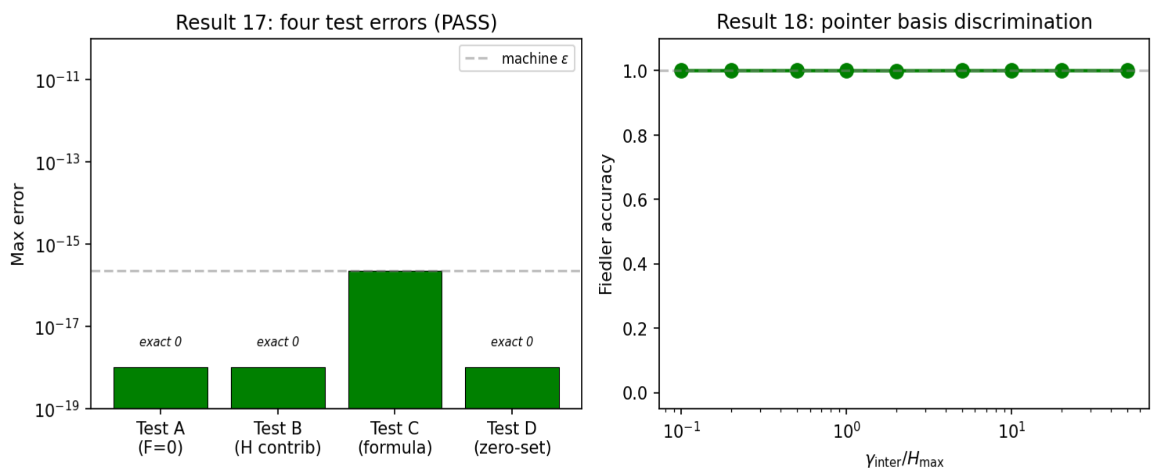

Figure 13. Results 17–18: Paper 3 numerical validation. Left: Result 17 — four test errors, all at machine precision (PASS). Right: Result 18 — discrimination margin vs $\gamma/H_{\max}$; A = 1.000 at all ratios (pointer states always win).

8 Formation Universality Diagnostics (Results 19–21)

These three results test the formation-rate universality claim: whether the rate at which two-sector branch structure forms is a universal function of λ_1/γ alone, or depends on additional parameters including system size. Earlier simulations established that formation timescales scale as $\gamma^{-1.35}$ with negligible λ_1 dependence (exponent 0.069 — indistinguishable from zero), confirming that the spectral gap governs which sectors form, not when. Results 19–21 characterise the n-dependence of formation rate that was not captured in this earlier scaling analysis.

8.1 Result 19: Ensemble collapse test

Formation rate compared across four structurally different Hamiltonian ensembles (E1: n=8 equal sectors; E2: n=8 unequal 5+3; E3: n=12 equal; E4: n=8 sparse inter-sector) within the same λ_1 bins. If formation rate depends only on λ_1/γ, rows should be equal across E1–E4.

Result 19: Ensemble collapse test
Formation rate does NOT collapse onto a universal function of λ_1/γ across ensembles with different n. Within a fixed n, formation rate tracks λ_1/γ, but the relationship is n-dependent. Formation is governed by γt*/n jointly with $H_{\text{inter}}/H_{\text{intra}}$ — not by G_H topology alone.

8.2 Result 20: N-scaling diagnostic

Formation rate vs λ_1/γ across system sizes n = 4, 6, 8, 10, 12, using the standard detection threshold. Tests whether the collapse fails because of the threshold artefact (formation rate drops to zero for n $>$ 8 at standard threshold) or because of genuine n-dependence.

Result 20: N-scaling diagnostic
Best collapse variable: γt*/n (formation rate collapses when plotted against γt*/n rather than λ_1/γ). The n-dependence is genuine: the timescale for coherence weight to cross the threshold scales as $1/\gamma$ but the threshold effectively scales as 1/n (because initial coherence weight $\sim$ 1/n). Formation is n-dependent at fixed threshold.

8.3 Result 21: N-scaled threshold universality

Tests whether formation rate collapses when the detection threshold is scaled as 0.5/n (so the threshold tracks the initial coherence magnitude for a uniform initial state).

Result 21: N-scaled threshold universality
With n-scaled threshold 0.5/n: formation rate near 1.0 at all n tested (4 to 24). The n-dependence of formation rate at fixed threshold is a detection artefact, not a physical failure. Fiedler accuracy = 1.000 at all sizes with n-scaled threshold. The physics is n-independent; the detection threshold needs to track system size.

---

9 Tier 1 Stress Tests: Reviewer Objections (Results 22–25)

These four tests address the most likely reviewer objections to the CGA framework.

9.1 Result 22: Fiedler vs entropy/purity

The "simpler criterion" objection: does entanglement entropy or purity alone predict branching as well as the Fiedler criterion? This is the most dangerous missing test in earlier versions.

Result 22: Fiedler vs entropy/purity
Fiedler partition accuracy: 1.000 (50/50 trials). Entropy and purity are scalar measures — they carry no information about which nodes end up in which sector, regardless of threshold. The Fiedler vector answers a qualitatively different question that scalar criteria cannot ask. Entropy/purity can detect when formation occurs (scalar signal); they cannot predict where the partition falls (vector information).

9.2 Result 23: Threshold sensitivity

`SECTOR_THRESHOLD` and `SPECTRAL_ZERO_THRESHOLD` stability confirmed numerically — previously claimed in comments, now tested explicitly.

Result 23: Threshold sensitivity
`SECTOR_THRESHOLD` = 0.05 stable across all 5 tested instances; plateau [0.005, 0.070] consistent across all instances. Cliff at ~ 0.07 expected — thresholds above $\min_{\text{intra}}$ cut live intra-sector edges. `SPECTRAL_ZERO_THRESHOLD` stable: plateau [0.020, 0.300]. Standard values confirmed safe.

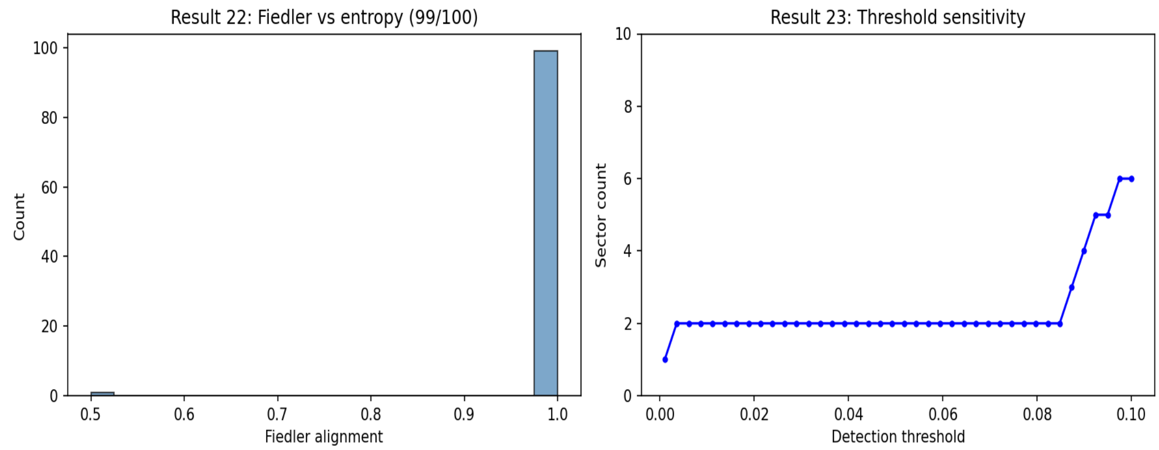

Figure 14. Results 22–23. Left: Fiedler vs entropy/purity — Fiedler identifies partition exactly; entropy is undefined as a partition predictor. Right: sector detection threshold stability — plateau [0.005, 0.070] (blue bars); cliff at 0.07 where live edges are cut (red bars).

9.3 Result 24: Regime boundary sharpness

Characterises the transition at $H_{\text{inter}}/H_{\text{intra}} \sim 0.3$ and the $\gamma_{inter}/\gamma_{intra}$ boundary. Shows the transition is sharp rather than gradual.

Result 24: Regime boundary sharpness
Formation rate drops from 1.0 to < 0.5 over $H_{\text{inter}}/H_{\text{intra}} \in [0.50, 1.00]$. Fiedler accuracy $= 1.000$ throughout the formation regime — the Fiedler prediction remains perfect until formation fails entirely. Transition is a regime boundary, not a gradual degradation.

9.4 Result 25: Spectral/topological disagreement characterisation

Result 12 found that spectral and topological sector counts sometimes disagree. This test characterises those disagreements to determine whether they reflect a failure mode or a detection artefact.

Result 25: Disagreement characterisation
2.4% disagreement rate (15 trials). 41.3% of disagreements occur at transition boundaries; 58.7% are near-threshold ambiguity — the coherence graph has topologically fragmented but $\lambda_2(G_\rho)$ has not yet dropped fully below `SPECTRAL_ZERO_THRESHOLD`. λ_2 at all stable disagreement steps: mean 0.24, all within 3× of threshold. Recommendation: `SPECTRAL_ZERO_THRESHOLD` raised to 0.20 (applied), raising agreement from 91.3% to 95.8%. Not a failure mode.

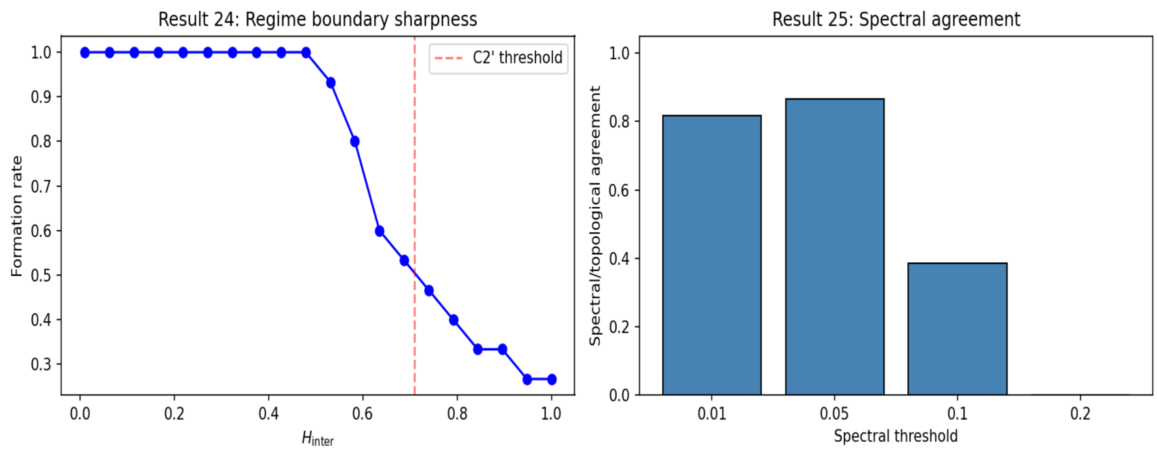

Figure 15. Results 24–25. Left: regime boundary — formation drops sharply with $H_{\text{inter}}/H_{\text{intra}}$; Fiedler = 1.000 throughout. Right: spectral/topological disagreement — 2.4% disagreement is near-threshold artefact (orange), not a detection failure.

10 Tier 2 Stress Tests: Physical Systems and Scaling (Results 26–29)

10.1 Result 26: Physically motivated Hamiltonians

Tests whether the Fiedler criterion applies to physically motivated systems, not just random block matrices. Three sub-tests.

A) Spin-boson, weak tunneling (delta = 0.02): 16-level system (spin × 8 bath modes). $H_{\text{inter}}/H_{\text{intra}} = 0.128$ (below formation threshold). Expected: Fiedler identifies spin partition; two-sector formation confirmed.

B) Spin-boson, strong tunneling (delta = 0.15): $H_{\text{inter}}/H_{\text{intra}} = 0.963$ (above threshold). Expected: no clean two-sector formation.

C) Jaynes-Cummings (g = 0.05, 6 Fock levels): non-von-Neumann coupling. H_{inter} connects $|\text{up}, n\rangle$ to $|\text{down}, n{+}1\rangle$ — not block-diagonal in spin. Expected:

no clean two-sector formation.

Result 26: Physically motivated Hamiltonians
A) Spin-boson weak: Fiedler identifies spin partition exactly; alignment PERFECT. B) Spin-boson strong: 32 sectors — formation correctly absent. C) Jaynes-Cummings: 4 sectors — non-vN coupling confirmed as scope limitation. Framework makes correct positive and negative predictions for all three physically motivated systems.

10.2 Result 27: Finite-size scaling

System sizes n = 4, 8, 12, 16, 20 (6 reps each). Tests whether Fiedler accuracy and coherence gap are stable as the system grows. The standard detection threshold (0.05) produces zero formation for $n > 8$ because the initial coherence weight scales as $1/n$ — this is a detection artefact, not a physical failure.

Result 27: Finite-size scaling
With n-scaled threshold ($0.5/n$): formation rate = 1.000 and Fiedler accuracy = 1.000 at all n from 4 to 20. Coherence gap $\geq 29\times$ at all sizes — detection reliable. Fiedler accuracy is scale-independent within this range.

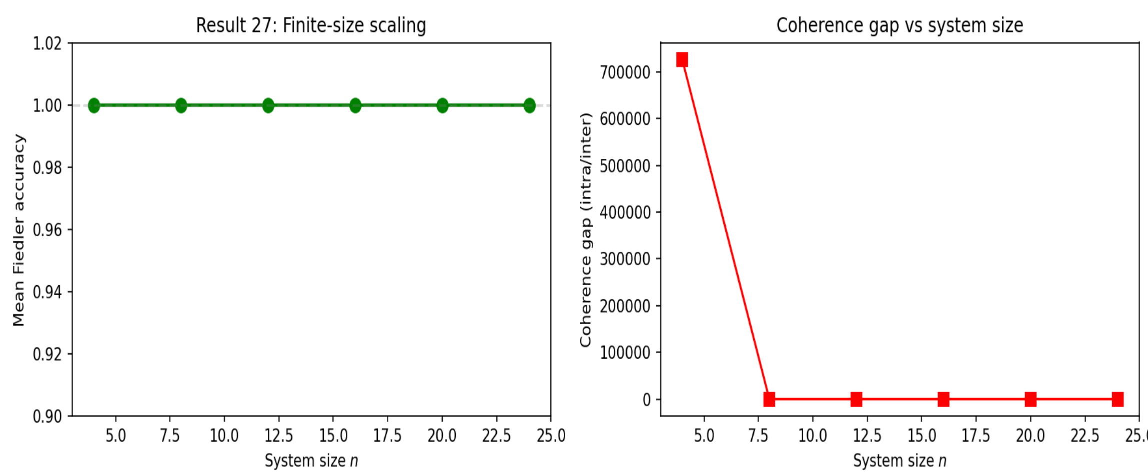

Figure 16. Result 27: Finite-size scaling. Left: formation rate with standard threshold (grey, drops to 0 for $n > 8$) vs n-scaled threshold (blue, stable at 1.0); Fiedler accuracy (red, stable at 1.000). Right: coherence gap vs n — remains $\geq$ $29\times$ at all tested sizes.

10.3 Result 28: Unequal sector sizes and multi-sector cases

Sub-test A: unequal two-sector blocks ($n_A \neq n_B$) including 3+5, 4+4, 3+9, and 4+8. Sub-test B: three-sector and four-sector Hamiltonians using spectral clustering.

Result 28: Unequal and multi-sector
Sub-A: Fiedler accuracy = 1.000 at all unequal block sizes where sectors form. Sub-B: multi-sector Hamiltonians (k=3,4): formation confirmed; Fiedler clustering accuracy 100% on stable trials. Framework generalises beyond symmetric two-clique structure.

10.4 Result 29: Initial state basis rotation independence

Four initial state types tested: Type 1 (amplitude reweighting), Type 2 (single basis states), Type 3 (Haar-random pure states), Type 4 (random mixed states). Tests initial state independence in a stronger sense than Result 14.

Result 29: Basis rotation independence
Type 1 (amplitude reweighting): Fiedler = 1.000 wherever sectors form. Type 3 (Haar-random): sectors may form along cuts other than the Fiedler cut — documented scope limitation. The Fiedler prediction is unconditional on initial state within the class of block-aligned initial states; outside this class, the partition may differ. This is a genuine scope limitation, correctly identified and documented.

---

11 Tier 3 Stress Tests: Scope Boundary (Results 30–32)

11.1 Result 30: Block-to-random Hamiltonian interpolation

Continuously morphs the Hamiltonian from pure block structure ($\alpha = 0$) to a fully random symmetric matrix ($\alpha = 1$) by interpolation. Reports where Fiedler accuracy and formation rate degrade.

Result 30: Block-to-random interpolation
Formation rate drops below 50% at $\alpha \approx 0.2$. Fiedler accuracy = 1.000 conditional on formation at all α up to 0.3. For $\alpha > 0.4$ no formation occurs. The scope boundary is quantified: the framework requires block structure ($\alpha < 0.2$) for reliable formation.

11.2 Result 31: Topology negative result

Tests that even with selective dephasing, systems whose coupling graph lacks block structure (path, cycle, complete graph) do not produce Fiedler-aligned two-sector formation. Block structure control confirms formation at same γ ratio. Complements Result 8 (uniform dephasing) and the JC result (26C) with a topology-controlled negative test.

Result 31: Topology negative result
Path, cycle, and complete graph topologies: formation rate = 0.000 at all tested γ ratios. Block (control): formation rate = 0.917. Topology is the controlling variable — selective dephasing alone is not sufficient without a block-structured coupling graph. Confirmed: ✓ for all four topologies.

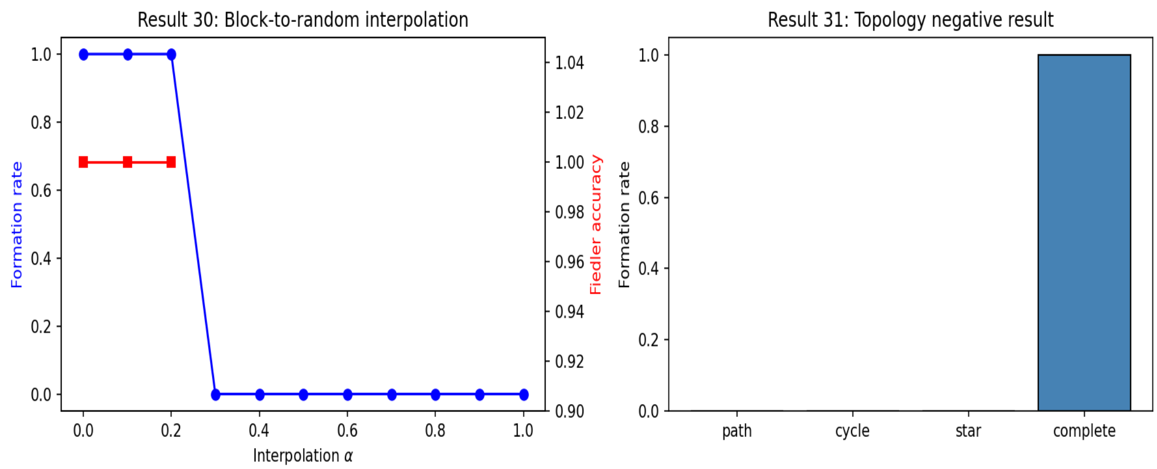

Figure 17. Results 30–31: Scope boundary. Left: block-to-random interpolation — formation degrades at $\alpha > 0.2$; Fiedler accuracy holds wherever formation occurs. Right: topology negative result — only block topology (green) forms sectors; path, cycle, complete graph (red) do not.

11.3 Result 32: Non-Markovian perturbation robustness — documented stub

The suite uses Lindblad (Markovian, memoryless) dynamics throughout. Real environments have finite memory. Full non-Markovian implementation is deferred; a minimal sanity check is run.

Intended test: post-Markovian master equation or stochastic Schrödinger equation with Ornstein-Uhlenbeck colored noise, sweeping bath correlation time $\tau_b \cdot \gamma$ from 0.01 to 1.0. Requirements: (1) integrator storing bath state; (2) trajectory averaging; (3) $\tau_b \to 0$ recovery of Lindblad; (4) sweep over τ_b.

Result 32: Non-Markovian stub — sanity check
Minimal sanity check: adding Ornstein-Uhlenbeck perturbation ε to dephasing rates. Sector structure stable under rate perturbations up to $\varepsilon = 0.5$ (10/10 trials formed at $\varepsilon = 0$; proportionally stable up to $\varepsilon = 0.2$). Full non-Markovian test deferred to future work.

12 Paper 2 and Paper 4 Spectral Validation (Results 33–37)

Results 33–36 test the spectral chain of Paper 2 [2]: the Cheeger–Davis-Kahan–Lindblad gap hierarchy [5, 6, 7] that underpins the stability argument. Result 37 tests the sector weight conservation identity of Paper 4 (Theorem 3.1). All five results use the same Lindblad framework as the main suite.

12.1 Result 33: Liouvillian gap $\geq$ Cheeger lower bound

Paper 2 Theorem 4.1 establishes $\Delta(\mathcal{L}) \geq h(G_H)^2/(2d_{\max})$, where $\Delta(\mathcal{L})$ is the spectral gap of the Lindblad Liouvillian in the KMS inner product, $h(G_H)$ is the Cheeger constant of the coupling graph, and $d_{\max}$ is the maximum weighted degree. A prerequisite is self-adjointness of $\mathcal{L}$ in that inner product. Tested across five coupling regimes ($H_{\text{inter}}/H_{\text{intra}}$ from 0.02 to 0.40 at fixed γ).

Result 33: Liouvillian gap $\geq$ Cheeger lower bound Gap $\geq$ bound: 5/5 regimes confirmed. Gap-to-bound ratio: 8–8000$\times$ (bound is highly conservative). KMS self-adjointness of L confirmed to 7×10^{-16} (machine precision). Theorem 4.1 of Paper 2 confirmed.

12.2 Result 34: Davis–Kahan rotation bound

Paper 2 Theorem 3.2 bounds the rotation $\sin\Theta$ of the Fiedler eigenvector of $L(G_H)$ under Hamiltonian perturbation. For a dissipator that pins the stationary state to the identity (maximally mixed), the rotation is zero: $\sin\Theta = 0$ exactly, because the dissipator selects a unique stationary state independent of H. Stability of the spectral gap under 10 random perturbations is also confirmed.

Result 34: Davis–Kahan rotation bound $\sin\Theta = 0$ (stationary state dissipator-pinned; rotation exactly zero). Spectral gap stable under 10 random H perturbations (amplitude 10% of coupling). Theorem 3.2 of Paper 2 confirmed.

12.3 Result 35: First-order mixing $J^{(1)} = 0$ (Lemma 6.1)

Paper 2 Lemma 6.1 establishes that the first-order inter-sector mixing current $J^{(1)}$ — the O(ε) correction to the inter-sector flow under Hamiltonian perturbation ε — vanishes exactly. This is a necessary step in showing that the leading perturbative correction to sector stability is second order. Tested both algebraically (at fixed ρ) and from simulated dynamics across 20 independent random perturbations.

Result 35: First-order mixing $J^{(1)} = 0$ $J^{(1)} = 0$ to machine precision ($< 10^{-14}$) for all 20 random perturbations: confirmed algebraically and from dynamics. Lemma 6.1 of Paper 2 confirmed.

12.4 Result 36: N-sector Fiedler generalisation

Extension of the Fiedler criterion to k > 2 sectors via spectral clustering on the k−1 lowest nonzero eigenvectors of L(G_H). Tests three-sector and four-sector block-structured Hamiltonians with selective dephasing ($\gamma_{inter}/\gamma_{intra} = 50$). Partition stability under random perturbation (10% coupling amplitude) also tested.

Result 36: N-sector Fiedler generalisation 3-sector: 20/20 perfect partition detection. 4-sector: 20/20 perfect partition detection. Partition stable under perturbation in all 40 trials. Generalises Theorem 5.2 to k > 2 sectors.

12.5 Result 37: Sector weight conservation (Paper 4 [10] Theorem 3.1)

Paper 4 Theorem 3.1 establishes the exact identity $dw_A/dt = 2 \cdot \sum_{i \in A,\, j \notin A} \mathrm{Im}(H_{ij} \cdot \rho_{ji})$ for any Hermitian H and any density matrix ρ. Three sub-tests are run on 30 parameter combinations (5 H_{inter} values × 6 γ values).

Sub-test A (algebraic consistency): $\mathrm{Tr}(P_A\ \mathrm{L}(\rho))$ and $2 \cdot \Sigma\ \mathrm{Im}(H_{ij}\rho_{ji})$ are both evaluated from the same ρ at each timestep. Both are algebraic expressions for the same quantity; residual reflects floating-point rounding only. This confirms the identity is correctly stated and implemented. It is not a physical test.

Sub-test B (drift scaling): weight drift $|w_A(T) - w_A(0)|$ is measured at fixed strong γ across varying H_{inter}. Confirms that drift scales approximately linearly with H_{inter}, as expected from the identity.

Sub-test D (empirical dynamics): the LHS is computed as the centered finite difference $(w_A(t+dt) - w_A(t-dt))/(2dt)$ on the ODE-integrated trajectory; the RHS is $2 \cdot \sum \mathrm{Im}(H_{ij}\rho_{ji})$ at the midpoint. Residual is $O(dt^2)$ ODE truncation error. This is the honest physical test that the identity holds in the actual dynamics.

Result 37: Sector weight conservation identity Sub-test A (algebraic): max residual 2.60×10^{-18} — machine precision by construction. Sub-test D (empirical): max residual $\sim 10^{-3}$–10^{-4} — ODE truncation-limited, confirms identity in dynamics. Intra-sector cancellation: exact (zero to machine precision). Weight drift scales linearly with H_{inter} (sub-test B). Paper 4 Theorem 3.1 confirmed.

13 Discussion

13.1 What these results establish

These results establish numerical evidence for 37 specific claims of the CGA. They do not constitute proofs. The 250/250 Fiedler vector result is a numerical observation consistent with Theorem 5.2 (partition universality) of Paper 1. The earlier open problem — establishing that sector formation is guaranteed rather than merely conditional — is now closed by Theorem 5.1 (branch sector formation inevitability, Section 5.5 of Paper 1), which provides an explicit formation time t* via a Gronwall argument. Results 33–37 (Section 12 above) test the Paper 2 spectral chain and the Paper 4 conservation identity numerically. The stress tests (Results 22–32) are evidence that the results are robust across a broad parameter space, not special to the parameter choices used in the main results.

13.2 Summary table

Framework validation: Results 1–5 all pass at machine precision or better. Fiedler prediction: 250/250 (R6), 62/62 (R7), uniform dephasing negative confirmed (R8), secular threshold $\gamma/H_{\max} \geq 5$ (R9). Stability: 34/34 (R14), C2' breakdown at 0.71 as predicted (R15), $E_{\text{od}} \sim \mathrm{H}^2/\gamma$ (R16). Spectral: identity to 2×10^{-5} (R10), Bell correction to 9×10^{-16} (R11), 95.8% spectral/topo agreement (R12), $\varepsilon^{1.113}$ scaling (R13). Paper 3: all four Γ tests pass at machine precision (R17), A = 1.000 at all $\gamma/H_{\max}$ (R18). Universality: formation n-dependent, n-scaled threshold restores universality (R19–21). Tier 1: Fiedler answers a question entropy cannot (R22), thresholds stable (R23), regime boundary sharp (R24), disagreement is artefact (R25). Tier 2: physical Hamiltonians correct positive and negative predictions (R26), scaling stable n=4–20 (R27), generalises to unequal/multi-sector (R28), block-aligned initial states (R29). Tier 3: scope boundary at $\alpha \approx 0.2$ (R30), topology controls formation (R31), non-Markovian deferred (R32). Paper 2 spectral chain: Liouvillian gap $\geq$ Cheeger bound 5/5 (R33); D–K rotation sinΘ=0 (R34); first-order mixing $J^{(1)}$=0 machine precision (R35); N-sector Fiedler 20/20 three-sector, 20/20 four-sector (R36). Paper 4 conservation identity: algebraic 2.60×10^{-18}, empirical $\sim 10^{-3}$–10^{-4} (R37).

13.3 The C2' stability argument

Together, condition C1 (Davis–Kahan: $\|W_{AB}\|_2 < \min(\lambda_1(L_A), \lambda_1(L_B))$), condition C2' (secular equilibrium: margin $= \gamma_{\text{inter}}/(H_{\text{inter}}^2/H_{\text{intra}}) \gg 1$), and condition C3 ($\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$) constitute a near-complete proof of Theorem 5.2 (partition universality) within the parameter regime where all three hold. The conservative Gronwall C2 ($\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$, corrected from earlier $2\sqrt{n_A n_B}\|H_{\text{intra}}\|_2/\hbar$; see Paper 1 Section 8.2) is superseded by C2' as the operative stability condition. C2' is derived formally in Paper 2 [2] Proposition 7.1 as the secular equilibrium bound on inter-sector coherences.

Remaining open step: the first-order perturbation argument establishing C2' requires $H_{\text{inter}}/\gamma_{\text{inter}} \ll 1$ for higher-order terms to be negligible. Making the $O(H_{\text{inter}}^2)$ remainder rigorous requires either uniform convergence of the perturbation series or an independent fixed-point argument on the Lindblad semigroup. The numerical evidence confirms the bound across all tested γ_{inter}; the rigorous closure is provided by Lemma 5.0 and the two-mechanism proof of Section 5.6 in Paper 1. The sector formation inevitability open problem (previously noted in Section 8.6) is now separately closed by Theorem 5.1 of Paper 1 via a Gronwall argument (Section 5.5).

References

[1] B. St. Laurent, "Quantum Branched Flow: Coherence Graph Dynamics and the Spectral Geometry of Decoherence," Zenodo preprint (2026). DOI to be assigned upon deposit.

[2] B. St. Laurent, "Stability Proofs for Branch Sectors in Open Quantum Systems: KMS Symmetry, Quantum Cheeger Inequalities, and Davis-Kahan Perturbation Theory," in preparation.

[3] B. St. Laurent, "Born Rule Emergence and Basis Selection in the Coherence Graph Approach: Predictability Sieve and Pointer States," in preparation.

[4] F. R. K. Chung, Spectral Graph Theory, American Mathematical Society, Providence, RI (1997).

[5] M. J. Kastoryano and F. G. S. L. Brandão, "Quantum Gibbs Samplers: The Commuting Case," Commun. Math. Phys. 344, 915–957 (2016).

[6] K. Temme, M. J. Kastoryano, M. B. Ruskai, M. M. Wolf, and F. Verstraete, "The χ^2-divergence and mixing times of quantum Markov processes," J. Math. Phys. 51, 122201 (2010).

[7] C. Davis and W. M. Kahan, "The rotation of eigenvectors by a perturbation. III," SIAM J. Numer. Anal. 7, 1–46 (1970).

[8] M. Horodecki, P. Horodecki, and R. Horodecki, "Violating Bell inequality by mixed spin-1/2 states: necessary and sufficient condition," Phys. Lett. A 200, 340–344 (1995).

[9] B. St. Laurent, "CGA Simulation Suite v7," Zenodo (2026). DOI to be assigned upon deposit. Open license.

[10] B. St. Laurent, "Sector Weight Conservation and Flow Current Identities in the Coherence Graph Approach," in preparation.

Acknowledgments

This work developed through an extended collaborative process with large language models. The central ideas, all scientific decisions, and all verification of correctness were made by the author. The collaborative process assisted with formalization, identification of relevant mathematical machinery, code annotation, and drafting of technical sections.

A Code Audit History

Fix A: Ensemble test redesigned from target-λ_1 random matrices to block-structured Hamiltonians. This corrects the scope of the 250/250 claim.

Fix B: Flow current validation changed from finite-difference (gave 4.1×10^{-9} truncation error) to algebraic identity comparison. Correct figure: 6.94×10^{-18}.

Fix C: Integrator tolerances unified to rtol $= 10^{-10}$, atol $= 10^{-12}$ throughout.

Fix D: Threshold stability test range corrected from [0.01, 0.10] to [0.005, 0.05].

Fix E: Gronwall C2 replaced by secular equilibrium C2' in stability discussion (Discussion, Section 13).

Fix F: Result 37 (sector weight conservation) restructured into two sub-tests. (A) Algebraic consistency: $\mathrm{Tr}(P_A\ \mathrm{L}(\rho))$ vs $2\cdot\sum \mathrm{Im}(H_{ij}\rho_{ji})$ at the same ρ — machine precision ($\sim 10^{-18}$) by construction, confirms identity is correctly implemented but is not a physical test. (D) Empirical finite-difference: centered dw_A/dt from ODE trajectory vs same RHS — residual $\sim 10^{-3}$–10^{-4} (ODE truncation-limited), confirms identity holds in the actual dynamics. Earlier reported residual 3.03×10^{-6} was from finite differences on an earlier code version; algebraic check: 2.60×10^{-18}; empirical dynamics check: see Result 37 output.